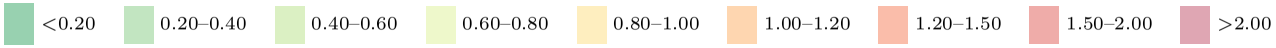
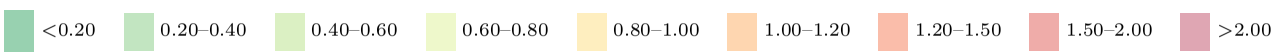
| | | Qwentaur | | | | | Llama-Centaur | | | Smoltaur | | | | Olmotaur | | [binz2025foundation] | | | | **Chance** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Experiment** | **Type** | 0.6B | 1.7B | 4B | 8B | 14B | 1B | 3B | 8B | 0.1B | 0.4B | 1.7B | 3B | 1B | 7B | L-70B$_p$ | C$_r$ | C$_p$ | Cog.$_p$ | ln($k$) |
| Balloon analog risk task [frey2017risk] | Decision | 0.07 | 0.07 | 0.07 | 0.06 | 0.06 | 0.07 | 0.07 | **0.06** | 0.08 | 0.07 | 0.07 | 0.07 | 0.07 | 0.07 | 0.08 | 0.07 | 0.06 | 0.09 | 0.69 |
| CPC18 [plonsky2018when] | Decision | 0.35 | 0.34 | 0.34 | 0.34 | **0.33** | 0.36 | 0.35 | 0.34 | 0.42 | 0.41 | 0.40 | 0.34 | 0.44 | 0.34 | 0.41 | 0.35 | 0.34 | 0.66 | 0.69 |
| Columbia card task [frey2017risk] | Decision | 0.20 | 0.20 | 0.20 | 0.20 | 0.20 | 0.21 | 0.20 | **0.20** | 0.24 | 0.22 | 0.21 | 0.20 | 0.22 | 0.20 | 0.23 | 0.21 | 0.19 | 0.26 | 0.69 |
| Decisions from description [wulff2018sampling] | Decision | 0.62 | 0.63 | 0.62 | 0.61 | 0.60 | 0.62 | 0.63 | 0.60 | 0.68 | 0.66 | 0.64 | 0.61 | 0.63 | 0.62 | 0.76 | **0.59** | 0.53 | 0.61 | 0.69 |
| Experiential-symbolic task [garcia2023experiential] | Decision | 0.48 | 0.47 | 0.46 | 0.46 | 0.46 | 0.47 | 0.47 | 0.46 | 0.47 | 0.45 | **0.43** | 0.47 | 0.47 | 0.47 | 0.70 | 0.46 | 0.45 | — | — |
| Gardening task [flesch2018comparing] | Decision | 0.39 | 0.39 | 0.38 | 0.38 | 0.38 | 0.40 | 0.39 | **0.38** | 0.57 | 0.54 | 0.53 | 0.38 | 0.58 | 0.38 | 0.50 | 0.49 | 0.38 | 0.91 | 0.69 |
| Go/no-go [enkavi2019large] | Decision | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 | **0.00** | 0.00 | 0.08 | 0.00 |
| Intertemporal choice [ruggeri2022globalizability] | Decision | 0.44 | 0.44 | 0.44 | 0.44 | 0.44 | 0.44 | 0.44 | 0.44 | 0.47 | 0.45 | 0.44 | 0.44 | 0.44 | 0.44 | 0.73 | **0.44** | 0.43 | 0.66 | 0.69 |
| Multi-attribute DM [hilbig2014generalized] | Decision | 0.09 | 0.08 | 0.09 | 0.08 | 0.08 | 0.08 | 0.08 | 0.08 | 0.31 | 0.19 | 0.16 | 0.08 | 0.10 | 0.09 | 0.15 | **0.06** | 0.06 | 0.19 | 0.69 |
| Risky choice [krueger2024identifying] | Decision | 0.48 | 0.47 | 0.44 | 0.43 | **0.42** | 0.49 | 0.46 | 0.43 | 0.72 | 0.61 | 0.53 | 0.45 | 0.58 | 0.46 | 0.65 | 0.43 | 0.43 | — | — |
| choices13k [peterson2021using] | Decision | 0.44 | 0.44 | 0.43 | 0.43 | **0.43** | 0.44 | 0.44 | 0.43 | 0.49 | 0.46 | 0.44 | 0.44 | 0.45 | 0.44 | 0.53 | 0.43 | 0.43 | 0.66 | 0.69 |
| Multi-task RL [tomov2021multitask] | MDP | 0.60 | 0.59 | 0.58 | 0.58 | **0.57** | 0.60 | 0.59 | 0.57 | 0.68 | 0.66 | 0.63 | 0.59 | 0.78 | 0.59 | 0.66 | 0.57 | 0.57 | 1.04 | 1.10 |
| Tile-revealing task [kumar2023disentangling] | MDP | 0.02 | 0.01 | 0.00 | 0.07 | 0.00 | 0.02 | 0.00 | **0.00** | 2.69 | 1.99 | 0.04 | 0.00 | 0.06 | 0.09 | 2.74 | 2.29 | 1.87 | — | — |
| Two-step task [kool2017cost] | MDP | 0.54 | 0.53 | **0.53** | 0.53 | **0.53** | 0.55 | 0.53 | **0.53** | 0.60 | 0.60 | 0.56 | 0.53 | 0.57 | 0.53 | 0.61 | 0.53 | 0.50$^\dagger$ | 0.60$^\dagger$ | 0.69 |
| Two-step task [kool2016when] | MDP | 0.51 | 0.50 | 0.49 | 0.49 | 0.48 | 0.52 | 0.50 | 0.49 | 0.59 | 0.58 | 0.52 | 0.50 | 0.55 | 0.50 | 0.61 | **0.48** | 0.50$^\dagger$ | 0.60$^\dagger$ | 0.69 |
| Two-step task [zorowitz2023data] | MDP | 0.52 | 0.51 | 0.51 | **0.51** | 0.51 | 0.52 | 0.51 | 0.51 | 0.60 | 0.58 | 0.52 | 0.52 | 0.54 | 0.51 | 0.61 | 0.51 | 0.50$^\dagger$ | 0.60$^\dagger$ | 0.69 |
| Virtual subway network [tomov2020discovery] | MDP | 1.16 | 1.15 | 1.14 | 1.13 | **1.12** | 1.19 | 1.16 | 1.13 | 1.43 | 1.33 | 1.30 | 1.14 | 1.32 | 1.15 | 1.53 | 1.16 | 1.13 | — | 1.61 |
| Zoopermarket [ludwig2023human] | MDP | 0.54 | 0.52 | 0.51 | 0.50 | **0.48** | 0.54 | 0.52 | 0.50 | 0.63 | 0.61 | 0.56 | 0.51 | 0.61 | 0.52 | 0.60 | 0.50 | 0.48 | 0.60 | 0.69 |
| Cond. assoc. learning [collins2014working] | Memory | 0.54 | 0.53 | 0.52 | 0.52 | **0.51** | 0.55 | 0.53 | 0.52 | 0.74 | 0.64 | 0.58 | 0.53 | 0.65 | 0.53 | 0.64 | 0.52 | 0.54 | 0.86 | 1.10 |
| Digit span [enkavi2019large] | Memory | 0.60 | 0.59 | 0.58 | 0.57 | **0.57** | 0.60 | 0.59 | 0.58 | 0.71 | 0.65 | 0.62 | 0.59 | 0.62 | 0.59 | 0.66 | 0.57 | 0.55 | 0.94 | 2.30 |
| Episodic long-term memory [popov2023intent] | Memory | 1.27 | 1.05 | 0.96 | 0.94 | 0.90 | 1.28 | 1.02 | 0.92 | 1.88 | 1.56 | 1.37 | 1.01 | 1.23 | 1.05 | 1.13 | **0.89** | 0.87 | — | 0.69 |
| N-back [enkavi2019large] | Memory | 0.42 | 0.42 | 0.41 | 0.41 | 0.40 | 0.42 | 0.42 | 0.41 | 0.49 | 0.47 | 0.44 | 0.41 | 0.44 | 0.42 | 0.52 | **0.40** | 0.40 | 0.58 | 0.69 |
| Recall and recognition [cox2018information] | Memory | 1.24 | 1.18 | 1.13 | 1.12 | 1.09 | 1.22 | 1.14 | 1.10 | 1.69 | 1.50 | 1.25 | 1.15 | 1.41 | 1.17 | 1.38 | **1.07** | 1.06 | — | — |
| Recent probes [enkavi2019large] | Memory | 0.26 | 0.26 | 0.26 | 0.26 | 0.26 | 0.27 | 0.26 | **0.26** | 0.33 | 0.28 | 0.27 | 0.26 | 0.27 | 0.26 | 0.34 | 0.26 | 0.26 | 0.39 | 0.69 |




**Bold+underline** = best; underline = second-best (C$_p$ excluded; see below). Lower is better. "—" = not available. All models finetuned at LoRA rank 16, bf16 inference. Reference columns ([binz2025foundation]): L-70B$_p$ = Llama-3.1-70B base, C$_r$ = our reproduced Centaur-70B, C$_p$ = published Centaur-70B, Cog.$_p$ = domain-specific cognitive baseline. C$_p$ is included for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. $^\dagger$Horizon and Two-step tasks are reported as single merged values by [binz2025foundation]. Rounded to 2 d.p.

Table 1: Per-experiment NLL on Psych-101 for all four finetuned families (bf16, LoRA rank 16). Cell colour encodes performance (green = better).

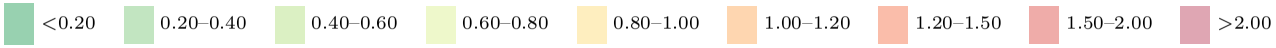
| | | Qwentaur | | | | | Llama-Centaur | | | Smoltaur | | | | Olmotaur | | [binz2025foundation] | | | | **Chance** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Experiment** | **Type** | 0.6B | 1.7B | 4B | 8B | 14B | 1B | 3B | 8B | 0.1B | 0.4B | 1.7B | 3B | 1B | 7B | L-70B$_p$ | C$_r$ | C$_p$ | Cog.$_p$ | ln($k$) |
| Grammar judgement [jansen2021rational] | Misc. | 1.47 | 1.45 | 1.45 | 1.44 | 1.43 | 1.48 | 1.45 | 1.44 | 1.63 | 1.54 | 1.51 | 1.46 | 1.51 | 1.46 | 1.99 | 1.44 | 1.44 | **1.41** | — |
| Probabilistic reasoning [zhu2020bayesian] | Misc. | 2.45 | 2.40 | 2.36 | 2.35 | **2.34** | 2.56 | 2.47 | 2.43 | 2.67 | 2.60 | 2.50 | 2.45 | 2.61 | 2.47 | 2.64 | 2.40 | 2.37 | — | — |
| Serial reaction time task [wu2023chunking] | Misc. | 0.17 | 0.17 | 0.17 | 0.17 | 0.17 | 0.17 | 0.17 | 0.17 | 0.17 | 0.16 | **0.16** | 0.17 | 0.16 | 0.17 | 0.19 | 0.18 | 0.17 | 0.20 | 1.39 |
| THINGS odd-one-out [hebart2023things] | Misc. | 0.82 | 0.81 | 0.80 | 0.79 | **0.79** | 0.82 | 0.82 | 0.79 | 1.08 | 0.90 | 0.86 | 0.80 | 0.86 | 0.81 | 1.14 | 0.80 | 0.81 | 0.83 | 1.10 |
| Changing bandit [xiong2023neural] | Bandit | 0.30 | 0.30 | 0.29 | 0.29 | **0.29** | 0.32 | 0.31 | 0.29 | 0.40 | 0.38 | 0.35 | 0.30 | 0.39 | 0.30 | 0.38 | 0.45 | 0.30 | 0.44 | 0.69 |
| Decisions from experience [wulff2018sampling] | Bandit | 0.49 | 0.49 | 0.49 | 0.49 | 0.48 | 0.50 | 0.49 | 0.48 | 0.54 | 0.52 | 0.50 | 0.49 | 0.50 | 0.49 | **0.43** | 0.47 | 0.37 | 0.54 | — |
| Drifting four-armed bandit [bahrami2020four] | Bandit | 0.73 | 0.72 | 0.71 | 0.71 | **0.71** | 0.73 | 0.72 | 0.71 | 0.83 | 0.78 | 0.74 | 0.72 | 0.74 | 0.72 | 0.88 | 0.71 | 0.70 | 0.90 | 1.39 |
| Horizon task [feng2021dynamics] | Bandit | 0.23 | 0.23 | 0.22 | 0.22 | 0.22 | 0.26 | 0.24 | **0.22** | 0.40 | 0.34 | 0.30 | 0.22 | 0.35 | 0.22 | 0.52 | 0.40 | 0.40$^\dagger$ | 0.36$^\dagger$ | 0.69 |
| Horizon task [sadeghiyeh2020temporal] | Bandit | 0.59 | 0.58 | 0.58 | 0.58 | 0.58 | 0.59 | 0.58 | 0.58 | 0.62 | 0.62 | 0.60 | 0.58 | 0.61 | 0.58 | 0.52 | 0.58 | 0.40$^\dagger$ | **0.36**$^\dagger$ | 0.69 |
| Horizon task [somerville2017charting] | Bandit | 0.34 | 0.34 | 0.34 | 0.34 | **0.33** | 0.35 | 0.34 | 0.34 | 0.48 | 0.40 | 0.38 | 0.34 | 0.40 | 0.34 | 0.52 | 0.35 | 0.40$^\dagger$ | 0.36$^\dagger$ | 0.69 |
| Horizon task [waltz2020differential] | Bandit | 0.15 | 0.15 | 0.15 | 0.15 | 0.15 | 0.16 | 0.16 | **0.14** | 0.26 | 0.21 | 0.18 | 0.15 | 0.19 | 0.15 | 0.52 | 0.15 | 0.40$^\dagger$ | 0.36$^\dagger$ | 0.69 |
| Horizon task [wilson2014humans] | Bandit | 0.46 | 0.46 | 0.45 | 0.45 | 0.45 | 0.47 | 0.46 | 0.45 | 0.54 | 0.51 | 0.49 | 0.45 | 0.52 | 0.45 | 0.52 | 0.48 | 0.40$^\dagger$ | **0.36**$^\dagger$ | 0.69 |
| Iowa gambling task [steingroever2015data] | Bandit | 0.93 | 0.93 | 0.91 | 0.91 | 0.91 | 0.94 | 0.92 | 0.91 | 1.01 | 0.98 | 0.94 | 0.92 | 0.95 | 0.92 | 0.99 | **0.91** | 0.89 | 1.16 | 1.39 |
| Prob. instrumental learning [lefebvre2017behavioural] | Bandit | 0.50 | 0.50 | **0.49** | 0.49 | 0.49 | 0.51 | 0.50 | 0.49 | 0.54 | 0.53 | 0.51 | 0.50 | 0.51 | 0.50 | 0.54 | 0.50 | 0.49 | 0.50 | 0.69 |
| Spatially correlated MAB [wu2018generalization] | Bandit | 2.01 | 1.96 | 1.93 | 1.87 | 1.86 | 2.16 | 2.00 | 1.93 | 2.50 | 2.30 | 2.16 | 1.95 | 2.26 | 2.06 | 2.45 | **1.82** | 1.83 | 2.76 | 3.40 |
| Structured bandit [schulz2020finding] | Bandit | 0.67 | 0.66 | 0.65 | 0.65 | **0.64** | 0.67 | 0.66 | 0.65 | 0.77 | 0.71 | 0.68 | 0.65 | 0.73 | 0.66 | 0.81 | 0.64 | 0.64 | 1.05 | 2.08 |
| Two-armed bandit [gershman2018deconstructing] | Bandit | 0.30 | 0.30 | 0.29 | 0.29 | **0.29** | 0.30 | 0.29 | 0.29 | 0.37 | 0.33 | 0.31 | 0.29 | 0.31 | 0.30 | 0.38 | 0.30 | 0.30 | 0.42 | 0.69 |
| Aversive learning [wise2019computational] | Sup. learn. | 4.71 | 4.55 | 4.27 | 4.24 | 3.81 | 4.58 | 4.80 | **3.66** | 5.35 | 5.12 | 4.92 | 3.93 | 5.44 | 4.67 | 5.11 | 4.13 | 4.07 | — | — |
| Medin categorization [levering2020revisiting] | Sup. learn. | 0.51 | 0.51 | 0.50 | 0.50 | **0.50** | 0.52 | 0.50 | 0.50 | 0.56 | 0.54 | 0.52 | 0.50 | 0.52 | 0.51 | 0.58 | 0.50 | 0.50 | 0.53 | — |
| Multiple-cue judgment [collsioo2023numerical] | Sup. learn. | 1.17 | 1.16 | 1.14 | 1.14 | 1.13 | 1.17 | 1.15 | **1.12** | 1.59 | 1.49 | 1.40 | 1.14 | 1.67 | 1.15 | 1.28 | 1.14 | 1.12 | 1.92 | 2.20 |
| Shepard categorization [badham2017deficits] | Sup. learn. | 0.55 | 0.55 | 0.54 | 0.53 | **0.53** | 0.56 | 0.54 | 0.53 | 0.62 | 0.58 | 0.56 | 0.54 | 0.56 | 0.54 | 0.58 | 0.54 | 0.54 | 0.61 | 0.69 |
| Weather prediction task [speekenbrink2008learning] | Sup. learn. | 0.55 | 0.56 | 0.54 | 0.55 | 0.55 | 0.55 | 0.54 | 0.56 | 0.58 | 0.56 | 0.56 | 0.55 | 0.57 | **0.53** | 0.57 | 0.56 | 0.55 | 0.63 | 0.69 |
| **Mean** | | 0.69 | 0.68 | 0.66 | 0.66 | **0.64** | 0.70 | 0.68 | 0.64 | 0.89 | 0.82 | 0.73 | 0.66 | 0.77 | 0.68 | 0.88 | 0.71 | 0.69 | 0.69 | 0.98 |




**Bold+underline** = best; underline = second-best (C$_p$ excluded; see below). Lower is better. "—" = not available. All models finetuned at LoRA rank 16, bf16 inference. Reference columns ([binz2025foundation]): L-70B$_p$ = Llama-3.1-70B base, C$_r$ = our reproduced Centaur-70B, C$_p$ = published Centaur-70B, Cog.$_p$ = domain-specific cognitive baseline. C$_p$ is included for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. $^\dagger$Horizon and Two-step tasks are reported as single merged values by [binz2025foundation]. Rounded to 2 d.p.

Table 1: Per-experiment NLL on Psych-101 for all four finetuned families (bf16, LoRA rank 16), continued. Cell colour encodes performance (green = better).

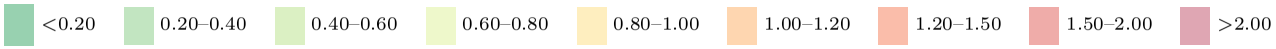
| | | Qwen3 family | | | | | | | | | | Llama-3.1/3.2 family | | | | | | [binz2025foundation] | | | | Chance |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 0.6B | | 1.7B | | 4B | | 8B | | 14B | | 1B | | 3B | | 8B | | | | | | |
| **Experiment** | **Type** | base | ft | base | ft | base | ft | base | ft | base | ft | base | ft | base | ft | base | ft | L-70B$_p$ | C$_r$ | C$_p$ | Cog.$_p$ | ln($k$) |
| Balloon analog risk task [frey2017risk] | Decision | 0.09 | 0.07 | 0.09 | 0.07 | 0.08 | 0.07 | 0.08 | 0.06 | 0.08 | 0.06 | 0.10 | 0.07 | 0.08 | 0.07 | 0.08 | **0.06** | 0.08 | 0.07 | 0.06 | 0.09 | 0.69 |
| CPC18 [plonsky2018when] | Decision | 0.41 | 0.35 | 0.40 | 0.34 | 0.40 | 0.34 | 0.40 | 0.34 | 0.40 | **0.33** | 0.43 | 0.36 | 0.41 | 0.35 | 0.39 | 0.34 | 0.41 | 0.35 | 0.34 | 0.66 | 0.69 |
| Columbia card task [frey2017risk] | Decision | 0.27 | 0.20 | 0.26 | 0.20 | 0.26 | 0.20 | 0.25 | 0.20 | 0.25 | 0.20 | 0.30 | 0.21 | 0.27 | 0.20 | 0.25 | **0.20** | 0.23 | 0.21 | 0.19 | 0.26 | 0.69 |
| Decisions from description [wulff2018sampling] | Decision | 1.08 | 0.62 | 1.05 | 0.63 | 1.01 | 0.62 | 0.99 | 0.61 | 1.01 | 0.60 | 1.55 | 0.62 | 1.20 | 0.63 | 0.96 | 0.60 | 0.76 | **0.59** | 0.53 | 0.61 | 0.69 |
| Experiential-symbolic task [garcia2023experiential] | Decision | 0.76 | 0.48 | 0.68 | 0.47 | 0.67 | 0.46 | 0.64 | 0.46 | 0.57 | 0.46 | 0.78 | 0.47 | 0.69 | 0.47 | 0.64 | 0.46 | 0.70 | **0.46** | 0.45 | — | — |
| Gardening task [flesch2018comparing] | Decision | 0.52 | 0.39 | 0.52 | 0.39 | 0.49 | 0.38 | 0.47 | 0.38 | 0.45 | 0.38 | 0.81 | 0.40 | 0.64 | 0.39 | 0.48 | **0.38** | 0.50 | 0.49 | 0.38 | 0.91 | 0.69 |
| Go/no-go [enkavi2019large] | Decision | 0.01 | **0.00** | 0.01 | **0.00** | 0.00 | **0.00** | 0.01 | **0.00** | 0.00 | **0.00** | 0.01 | **0.00** | 0.01 | **0.00** | 0.00 | **0.00** | 0.01 | **0.00** | 0.00 | 0.08 | 0.00 |
| Intertemporal choice [ruggeri2022globalizability] | Decision | 0.77 | 0.44 | 0.80 | 0.44 | 0.71 | 0.44 | 0.69 | 0.44 | 0.70 | 0.44 | 0.85 | 0.44 | 0.75 | 0.44 | 0.76 | 0.44 | 0.73 | **0.44** | 0.43 | 0.66 | 0.69 |
| Multi-attribute DM [hilbig2014generalized] | Decision | 0.49 | 0.09 | 0.34 | 0.08 | 0.20 | 0.09 | 0.19 | 0.08 | 0.19 | 0.08 | 0.63 | 0.08 | 0.45 | 0.08 | 0.23 | 0.08 | 0.15 | **0.06** | 0.06 | 0.19 | 0.69 |
| Risky choice [krueger2024identifying] | Decision | 0.92 | 0.48 | 0.83 | 0.47 | 0.79 | 0.44 | 0.74 | 0.43 | 0.67 | **0.42** | 0.99 | 0.49 | 0.85 | 0.46 | 0.76 | 0.43 | 0.65 | 0.43 | 0.43 | — | — |
| choices13k [peterson2021using] | Decision | 0.62 | 0.44 | 0.60 | 0.44 | 0.59 | 0.43 | 0.59 | 0.43 | 0.55 | **0.43** | 0.67 | 0.44 | 0.59 | 0.44 | 0.55 | 0.43 | 0.53 | 0.43 | 0.43 | 0.66 | 0.69 |
| Multi-task RL [tomov2021multitask] | MDP | 0.70 | 0.60 | 0.72 | 0.59 | 0.68 | 0.58 | 0.70 | 0.58 | 0.68 | **0.57** | 0.80 | 0.60 | 0.70 | 0.59 | 0.66 | 0.57 | 0.66 | 0.57 | 0.57 | 1.04 | 1.10 |
| Tile-revealing task [kumar2023disentangling] | MDP | 2.95 | 0.02 | 2.81 | 0.01 | 2.67 | 0.00 | 2.73 | 0.07 | 2.60 | 0.00 | 4.86 | 0.02 | 3.98 | 0.00 | 2.92 | **0.00** | 2.74 | 2.29 | 1.87 | — | — |
| Two-step task [kool2017cost] | MDP | 0.67 | 0.54 | 0.64 | 0.53 | 0.64 | **0.53** | 0.65 | 0.53 | 0.63 | **0.53** | 0.67 | 0.55 | 0.62 | 0.53 | 0.61 | **0.53** | 0.61 | 0.53 | 0.50$^{\dagger}$ | 0.60$^{\dagger}$ | 0.69 |
| Two-step task [kool2016when] | MDP | 0.69 | 0.51 | 0.64 | 0.50 | 0.64 | 0.49 | 0.65 | 0.49 | 0.63 | 0.48 | 0.67 | 0.52 | 0.62 | 0.50 | 0.61 | 0.49 | 0.61 | **0.48** | 0.50$^{\dagger}$ | 0.60$^{\dagger}$ | 0.69 |
| Two-step task [zorowitz2023data] | MDP | 0.62 | 0.52 | 0.59 | 0.51 | 0.60 | 0.51 | 0.61 | **0.51** | 0.59 | 0.51 | 0.64 | 0.52 | 0.60 | 0.51 | 0.58 | 0.51 | 0.61 | 0.51 | 0.50$^{\dagger}$ | 0.60$^{\dagger}$ | 0.69 |
| Virtual subway network [tomov2020discovery] | MDP | 1.68 | 1.16 | 1.55 | 1.15 | 1.57 | 1.14 | 1.55 | 1.13 | 1.57 | **1.12** | 1.68 | 1.19 | 1.60 | 1.16 | 1.53 | 1.13 | 1.53 | 1.16 | 1.13 | — | 1.61 |
| Zoopermarket [ludwig2023human] | MDP | 0.61 | 0.54 | 0.61 | 0.52 | 0.60 | 0.51 | 0.60 | 0.50 | 0.59 | **0.48** | 0.64 | 0.54 | 0.62 | 0.52 | 0.60 | 0.50 | 0.60 | 0.50 | 0.48 | 0.60 | 0.69 |
| Cond. assoc. learning [collins2014working] | Memory | 0.76 | 0.54 | 0.71 | 0.53 | 0.71 | 0.52 | 0.69 | 0.52 | 0.61 | **0.51** | 0.82 | 0.55 | 0.73 | 0.53 | 0.70 | 0.52 | 0.64 | 0.52 | 0.54 | 0.86 | 1.10 |
| Digit span [enkavi2019large] | Memory | 0.94 | 0.60 | 0.89 | 0.59 | 0.72 | 0.58 | 0.70 | 0.57 | 0.69 | **0.57** | 1.01 | 0.60 | 0.84 | 0.59 | 0.75 | 0.58 | 0.66 | 0.57 | 0.55 | 0.94 | 2.30 |
| Episodic long-term memory [popov2023intent] | Memory | 1.72 | 1.27 | 1.45 | 1.05 | 1.33 | 0.96 | 1.30 | 0.94 | 1.18 | 0.90 | 1.97 | 1.28 | 1.65 | 1.02 | 1.43 | 0.92 | 1.13 | **0.89** | 0.87 | — | 0.69 |
| N-back [enkavi2019large] | Memory | 0.58 | 0.42 | 0.58 | 0.42 | 0.55 | 0.41 | 0.55 | 0.41 | 0.53 | 0.40 | 0.59 | 0.42 | 0.57 | 0.42 | 0.57 | 0.41 | 0.52 | **0.40** | 0.40 | 0.58 | 0.69 |
| Recall and recognition [cox2018information] | Memory | 1.73 | 1.24 | 1.57 | 1.18 | 1.44 | 1.13 | 1.41 | 1.12 | 1.36 | 1.09 | 1.87 | 1.22 | 1.56 | 1.14 | 1.37 | 1.10 | 1.38 | **1.07** | 1.06 | — | — |
| Recent probes [enkavi2019large] | Memory | 0.70 | 0.26 | 0.61 | 0.26 | 0.39 | 0.26 | 0.41 | 0.26 | 0.35 | 0.26 | 0.74 | 0.27 | 0.56 | 0.26 | 0.34 | **0.26** | 0.34 | 0.26 | 0.26 | 0.39 | 0.69 |




**Bold+underline** = best; underline = second-best (C$_p$ excluded; see footnote). NLL under bf16 (half-precision) inference. Reference columns report 4-bit values as originally published. Lower is better. "—" = no cognitive baseline reported. Size labels in billions of parameters. Subscript $_r$ denotes our reproducing evaluation of the original Centaur model under identical library and CUDA versions as our models; $_p$ denotes values published by [binz2025foundation]. C$_p$ is included for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. $^{\dagger}$Horizon task (5 experiments) and Two-step task (3 experiments) are reported as single merged values by [binz2025foundation]. Rounded to 2 d.p.

Table 2: Base vs. finetuned NLL on Psych-101 under bf16 inference. Within each model size, the left column (base) shows the pretrained model and the right column (ft) shows the cognitively finetuned variant. Cell colour encodes performance; the consistent colour shift from base to ft demonstrates the effect of cognitive finetuning across all scales.

| | | Qwen3 family | | | | | | | | | | Llama-3.1/3.2 family | | | | | | [binz2025foundation] | | | | Chance |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 0.6B | | 1.7B | | 4B | | 8B | | 14B | | 1B | | 3B | | 8B | | | | | | |
| **Experiment** | **Type** | base | ft | base | ft | base | ft | base | ft | base | ft | base | ft | base | ft | base | ft | L-70B$_p$ | C$_r$ | C$_p$ | Cog.$_p$ | ln($k$) |
| Grammar judgement [jansen2021rational] | Misc. | 2.12 | 1.47 | 2.02 | 1.45 | 2.00 | 1.45 | 1.89 | 1.44 | 1.91 | 1.43 | 2.33 | 1.48 | 2.16 | 1.45 | 2.12 | 1.44 | 1.99 | 1.44 | 1.44 | **1.41** | — |
| Probabilistic reasoning [zhu2020bayesian] | Misc. | 2.75 | 2.45 | 2.69 | 2.40 | 2.61 | 2.36 | 2.57 | 2.35 | 2.58 | **2.34** | 2.89 | 2.56 | 2.80 | 2.47 | 2.71 | 2.43 | 2.64 | 2.40 | 2.37 | — | — |
| Serial reaction time task [wu2023chunking] | Misc. | 0.21 | 0.17 | 0.19 | 0.17 | 0.19 | 0.17 | 0.19 | 0.17 | 0.19 | **0.17** | 0.31 | 0.17 | 0.23 | 0.17 | 0.20 | 0.17 | 0.19 | 0.18 | 0.17 | 0.20 | 1.39 |
| THINGS odd-one-out [hebart2023things] | Misc. | 1.21 | 0.82 | 1.21 | 0.81 | 1.17 | 0.80 | 1.15 | 0.79 | 1.13 | **0.79** | 1.31 | 0.82 | 1.18 | 0.82 | 1.16 | 0.79 | 1.14 | 0.80 | 0.81 | 0.83 | 1.10 |
| Changing bandit [xiong2023neural] | Bandit | 0.42 | 0.30 | 0.38 | 0.30 | 0.36 | 0.29 | 0.35 | 0.29 | 0.34 | **0.29** | 1.34 | 0.32 | 0.90 | 0.31 | 0.39 | 0.29 | 0.38 | 0.45 | 0.30 | 0.44 | 0.69 |
| Decisions from experience [wulff2018sampling] | Bandit | 0.74 | 0.49 | 0.66 | 0.49 | 0.69 | 0.49 | 0.78 | 0.49 | 0.61 | 0.48 | 0.81 | 0.50 | 0.71 | 0.49 | 0.65 | 0.48 | **0.43** | 0.47 | 0.37 | 0.54 | — |
| Drifting four-armed bandit [bahrami2020four] | Bandit | 0.98 | 0.73 | 0.95 | 0.72 | 0.92 | 0.71 | 0.91 | 0.71 | 0.91 | **0.71** | 0.95 | 0.73 | 0.94 | 0.72 | 0.90 | 0.71 | 0.88 | 0.71 | 0.70 | 0.90 | 1.39 |
| Horizon task [feng2021dynamics] | Bandit | 0.47 | 0.23 | 0.43 | 0.23 | 0.39 | 0.22 | 0.36 | 0.22 | 0.34 | 0.22 | 0.81 | 0.26 | 0.67 | 0.24 | 0.44 | **0.22** | 0.52 | 0.40 | 0.40$^\dagger$ | 0.36$^\dagger$ | 0.69 |
| Horizon task [sadeghiyeh2020temporal] | Bandit | 0.67 | 0.59 | 0.72 | 0.58 | 0.66 | 0.58 | 0.65 | 0.58 | 0.65 | 0.58 | 0.67 | 0.59 | 0.68 | 0.58 | 0.66 | 0.58 | 0.52 | 0.58 | 0.40$^\dagger$ | **0.36**$^\dagger$ | 0.69 |
| Horizon task [somerville2017charting] | Bandit | 0.54 | 0.34 | 0.55 | 0.34 | 0.53 | 0.34 | 0.47 | 0.34 | 0.49 | **0.33** | 0.59 | 0.35 | 0.57 | 0.34 | 0.51 | 0.34 | 0.52 | 0.35 | 0.40$^\dagger$ | 0.36$^\dagger$ | 0.69 |
| Horizon task [waltz2020differential] | Bandit | 0.33 | 0.15 | 0.33 | 0.15 | 0.31 | 0.15 | 0.27 | 0.15 | 0.24 | 0.15 | 0.37 | 0.16 | 0.37 | 0.16 | 0.30 | **0.14** | 0.52 | 0.15 | 0.40$^\dagger$ | 0.36$^\dagger$ | 0.69 |
| Horizon task [wilson2014humans] | Bandit | 0.57 | 0.46 | 0.58 | 0.46 | 0.56 | 0.45 | 0.53 | 0.45 | 0.54 | 0.45 | 0.64 | 0.47 | 0.62 | 0.46 | 0.55 | 0.45 | 0.52 | 0.48 | 0.40$^\dagger$ | **0.36**$^\dagger$ | 0.69 |
| Iowa gambling task [steingroever2015data] | Bandit | 1.07 | 0.93 | 1.06 | 0.93 | 1.07 | 0.91 | 1.07 | 0.91 | 1.07 | 0.91 | 1.10 | 0.94 | 1.06 | 0.92 | 1.04 | 0.91 | 0.99 | **0.91** | 0.89 | 1.16 | 1.39 |
| Prob. instrumental learning [lefebvre2017behavioural] | Bandit | 0.62 | 0.50 | 0.60 | 0.50 | 0.57 | **0.49** | 0.59 | 0.49 | 0.56 | 0.49 | 0.62 | 0.51 | 0.58 | 0.50 | 0.56 | 0.49 | 0.54 | 0.50 | 0.49 | 0.50 | 0.69 |
| Spatially correlated MAB [wu2018generalization] | Bandit | 2.61 | 2.01 | 2.55 | 1.96 | 2.50 | 1.93 | 2.44 | 1.87 | 2.46 | 1.86 | 2.76 | 2.16 | 2.64 | 2.00 | 2.58 | 1.93 | 2.45 | **1.82** | 1.83 | 2.76 | 3.40 |
| Structured bandit [schulz2020finding] | Bandit | 0.92 | 0.67 | 0.92 | 0.66 | 0.88 | 0.65 | 0.87 | 0.65 | 0.83 | **0.64** | 0.94 | 0.67 | 0.89 | 0.66 | 0.85 | 0.65 | 0.81 | 0.64 | 0.64 | 1.05 | 2.08 |
| Two-armed bandit [gershman2018deconstructing] | Bandit | 0.43 | 0.30 | 0.41 | 0.30 | 0.40 | 0.29 | 0.39 | 0.29 | 0.39 | **0.29** | 0.43 | 0.30 | 0.43 | 0.29 | 0.40 | 0.29 | 0.38 | 0.30 | 0.30 | 0.42 | 0.69 |
| Aversive learning [wise2019computational] | Sup. learn. | 5.20 | 4.71 | 5.06 | 4.55 | 4.95 | 4.27 | 4.89 | 4.24 | 4.75 | 3.81 | 5.66 | 4.58 | 5.52 | 4.80 | 5.25 | **3.66** | 5.11 | 4.13 | 4.07 | — | — |
| Medin categorization [levering2020revisiting] | Sup. learn. | 0.60 | 0.51 | 0.60 | 0.51 | 0.60 | 0.50 | 0.59 | 0.50 | 0.55 | **0.50** | 0.71 | 0.52 | 0.60 | 0.50 | 0.57 | 0.50 | 0.58 | 0.50 | 0.50 | 0.53 | — |
| Multiple-cue judgment [collsioo2023numerical] | Sup. learn. | 1.46 | 1.17 | 1.38 | 1.16 | 1.28 | 1.14 | 1.30 | 1.14 | 1.26 | 1.13 | 1.69 | 1.17 | 1.50 | 1.15 | 1.32 | **1.12** | 1.28 | 1.14 | 1.12 | 1.92 | 2.20 |
| Shepard categorization [badham2017deficits] | Sup. learn. | 0.63 | 0.55 | 0.61 | 0.55 | 0.59 | 0.54 | 0.58 | 0.53 | 0.59 | **0.53** | 0.67 | 0.56 | 0.62 | 0.54 | 0.60 | 0.53 | 0.58 | 0.54 | 0.54 | 0.61 | 0.69 |
| Weather prediction task [speekenbrink2008learning] | Sup. learn. | 0.62 | 0.55 | 0.59 | 0.56 | 0.61 | 0.54 | 0.58 | 0.55 | 0.58 | 0.55 | 0.62 | 0.55 | 0.59 | **0.54** | 0.58 | 0.56 | 0.57 | 0.56 | 0.55 | 0.63 | 0.69 |
| **Mean (all 46)** | | 0.99 | 0.69 | 0.94 | 0.68 | 0.90 | 0.66 | 0.89 | 0.66 | 0.87 | **0.64** | 1.14 | 0.70 | 1.02 | 0.68 | 0.92 | 0.64 | 0.88 | 0.71 | 0.69 | 0.69 | 0.98 |




**Bold+underline** = best; underline = second-best (C$_p$ excluded; see footnote). NLL under bf16 (half-precision) inference. Reference columns report 4-bit values as originally published. Lower is better. "—" = no cognitive baseline reported. Size labels in billions of parameters. Subscript $_r$ denotes our reproducing evaluation of the original Centaur model under identical library and CUDA versions as our models; $_p$ denotes values published by [binz2025foundation]. C$_p$ is included for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. $^\dagger$Horizon task (5 experiments) and Two-step task (3 experiments) are reported as single merged values by [binz2025foundation]. Rounded to 2 d.p.

Table 2: Base vs. finetuned NLL on Psych-101 under bf16 inference, continued. Within each model size, the left column (base) shows the pretrained model and the right column (ft) the cognitively finetuned variant.

| | Qwentaur | | | | | Llama-Centaur | | | Smoltaur | | | | Olmotaur | | Base | | | [binz2025foundation] | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Task type** | 0.6B | 1.7B | 4B | 8B | 14B | 1B | 3B | 8B | 0.1B | 0.4B | 1.7B | 3B | 1B | 7B | Q-8B | Q-14B | L-8B | $70B_p$ | $70B_r$ | $Cog._p$ |
| Decision (11) | 0.32 | 0.32 | 0.32 | 0.31 | **0.31** | 0.32 | 0.32 | 0.31 | 0.40 | 0.37 | 0.35 | 0.32 | 0.36 | 0.32 | 0.46 | 0.44 | 0.46 | 0.30 | 0.32 | 0.46 |
| MDP (7) | 0.56 | 0.55 | 0.54 | 0.54 | **0.53** | 0.56 | 0.54 | 0.53 | 1.03 | 0.91 | 0.59 | 0.54 | 0.63 | 0.56 | 1.07 | 1.04 | 1.07 | 0.79 | 0.86 | 0.69 |
| Bandit (13) | 0.59 | 0.59 | 0.58 | 0.57 | **0.57** | 0.61 | 0.59 | 0.58 | 0.71 | 0.66 | 0.63 | 0.58 | 0.65 | 0.59 | 0.74 | 0.73 | 0.76 | 0.58 | 0.60 | 0.74 |
| Memory (6) | 0.72 | 0.67 | 0.65 | 0.64 | 0.62 | 0.72 | 0.66 | 0.63 | 0.97 | 0.85 | 0.76 | 0.66 | 0.77 | 0.67 | 0.84 | 0.78 | 0.86 | 0.61 | **0.62** | 0.69 |
| Misc. (4) | 1.23 | 1.21 | 1.19 | 1.19 | 1.18 | 1.26 | 1.23 | 1.21 | 1.39 | 1.30 | 1.25 | 1.22 | 1.28 | 1.23 | 1.45 | 1.45 | 1.55 | 1.20 | 1.20 | **0.81** |
| Sup. learn. (5) | 1.50 | 1.47 | 1.40 | 1.39 | 1.30 | 1.48 | 1.51 | 1.27 | 1.74 | 1.66 | 1.59 | 1.33 | 1.75 | 1.48 | 1.59 | 1.55 | 1.66 | 1.36 | 1.37 | **0.92** |
| **Mean (46)** | 0.69 | 0.68 | 0.66 | 0.66 | **0.64** | 0.70 | 0.68 | 0.64 | 0.89 | 0.82 | 0.73 | 0.66 | 0.77 | 0.68 | 0.89 | 0.87 | 0.92 | 0.69 | 0.71 | 0.69 |

<0.20 0.20–0.40 0.40–0.60 0.60–0.80 0.80–1.00 >1.00

Table 3: Mean NLL by task type for finetuned models (bf16). Finetuned families: `Qwentaur`, `Llama-Centaur`, `Smoltaur`, `Olmotaur`. Base columns: `Q-8B`/`Q-14B` (Qwen3), `L-8B` (Llama-3.1). Subscript $_r$ denotes our reproducing evaluation of the original Centaur model distributed by [binz2025foundation], evaluated under identical python library and CUDA versions as our small foundation models for fair comparison. Subscript $_p$ denotes values published by [binz2025foundation]. $70B_p$ is shown for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. Cell colour encodes performance; **bold+underline** marks the best model, underline the second-best. Full per-experiment results in Appendix **??**.

| | Qwentaur | | | | | Llama-Centaur | | | Smoltaur | | | | Olmotaur | | Base | | | [binz2025foundation] | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Task type** | 0.6B | 1.7B | 4B | 8B | 14B | 1B | 3B | 8B | 0.1B | 0.4B | 1.7B | 3B | 1B | 7B | Q-8B | Q-14B | L-8B | $70B_p$ | $70B_r$ | $Cog._p$ |
| Decision (9) | 0.29 | 0.29 | 0.29 | 0.28 | **0.28** | 0.29 | 0.29 | 0.28 | 0.36 | 0.33 | 0.32 | 0.28 | 0.33 | 0.29 | 0.41 | 0.40 | 0.41 | 0.27 | 0.29 | 0.46 |
| MDP (5) | 0.54 | 0.53 | 0.52 | 0.52 | **0.51** | 0.54 | 0.53 | 0.52 | 0.62 | 0.60 | 0.56 | 0.53 | 0.61 | 0.53 | 0.64 | 0.62 | 0.61 | 0.51 | 0.52 | 0.69 |
| Bandit (13) | 0.59 | 0.59 | 0.58 | 0.57 | **0.57** | 0.61 | 0.59 | 0.58 | 0.71 | 0.66 | 0.63 | 0.58 | 0.65 | 0.59 | 0.74 | 0.73 | 0.76 | 0.58 | 0.60 | 0.74 |
| Memory (4) | 0.45 | 0.45 | 0.44 | 0.44 | **0.44** | 0.46 | 0.45 | 0.44 | 0.57 | 0.51 | 0.48 | 0.45 | 0.49 | 0.45 | 0.59 | 0.54 | 0.59 | 0.44 | 0.44 | 0.69 |
| Misc. (3) | 0.82 | 0.81 | 0.80 | 0.80 | **0.80** | 0.83 | 0.81 | 0.80 | 0.96 | 0.87 | 0.84 | 0.81 | 0.84 | 0.81 | 1.08 | 1.08 | 1.16 | 0.80 | 0.81 | 0.81 |
| Sup. learn. (4) | 0.69 | 0.70 | 0.68 | 0.68 | **0.68** | 0.70 | 0.68 | 0.68 | 0.84 | 0.79 | 0.76 | 0.68 | 0.83 | 0.68 | 0.77 | 0.75 | 0.77 | 0.68 | 0.68 | 0.92 |
| **Mean (38)** | 0.53 | 0.52 | 0.52 | 0.51 | **0.51** | 0.54 | 0.52 | 0.51 | 0.64 | 0.59 | 0.56 | 0.52 | 0.59 | 0.52 | 0.66 | 0.65 | 0.67 | 0.51 | 0.52 | 0.69 |

<0.20 0.20–0.40 0.40–0.60 0.60–0.80 0.80–1.00 >1.00

Table 4: Mean NLL by task type for finetuned models (bf16), restricted to the 38 experiments (of 46) for which a cognitive model baseline is available. 8 experiments without a reported cognitive baseline are excluded to enable direct comparison across all columns. Finetuned families: `Qwentaur`, `Llama-Centaur`, `Smoltaur`, `Olmotaur`. Base columns: `Q-8B`/`Q-14B` (Qwen3), `L-8B` (Llama-3.1). Subscript $_r$ denotes our reproducing evaluation of the original Centaur model distributed by [binz2025foundation], evaluated under identical python library and CUDA versions as our small foundation models for fair comparison. Subscript $_p$ denotes values published by [binz2025foundation]. $70B_p$ is shown for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. Cell colour encodes performance; **bold+underline** marks the best model, underline the second-best. Full per-experiment results in Appendix **??**.

| | Qwentaur | | | | | Llama-Centaur | | | Smoltaur | | | | Olmotaur | | Base | | | [binz2025foundation] | | | **Chance** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Task type** | 0.6B | 1.7B | 4B | 8B | 14B | 1B | 3B | 8B | 0.1B | 0.4B | 1.7B | 3B | 1B | 7B | Q-8B | Q-14B | L-8B | $70B_p$ | $70B_r$ | $Cog._p$ | $\ln(k)$ |
| Decision (9) | 0.29 | 0.29 | 0.29 | 0.28 | **0.28** | 0.29 | 0.29 | 0.28 | 0.36 | 0.33 | 0.32 | 0.28 | 0.33 | 0.29 | 0.41 | 0.40 | 0.41 | 0.27 | 0.29 | 0.46 | 0.62 |
| MDP (5) | 0.54 | 0.53 | 0.52 | 0.52 | **0.51** | 0.54 | 0.53 | 0.52 | 0.62 | 0.60 | 0.56 | 0.53 | 0.61 | 0.53 | 0.64 | 0.62 | 0.61 | 0.51 | 0.52 | 0.69 | 0.77 |
| Bandit (12) | 0.60 | 0.60 | 0.59 | 0.58 | **0.58** | 0.62 | 0.60 | 0.58 | 0.73 | 0.68 | 0.64 | 0.59 | 0.66 | 0.60 | 0.74 | 0.74 | 0.76 | 0.60 | 0.61 | 0.75 | 1.15 |
| Memory (4) | 0.45 | 0.45 | 0.44 | 0.44 | **0.44** | 0.46 | 0.45 | 0.44 | 0.57 | 0.51 | 0.48 | 0.45 | 0.49 | 0.45 | 0.59 | 0.54 | 0.59 | 0.44 | 0.44 | 0.69 | 1.20 |
| Misc. (2) | 0.50 | 0.49 | 0.48 | 0.48 | **0.48** | 0.50 | 0.49 | 0.48 | 0.62 | 0.53 | 0.51 | 0.48 | 0.51 | 0.49 | 0.67 | 0.66 | 0.68 | 0.49 | 0.49 | 0.51 | 1.24 |
| Sup. learn. (3) | 0.75 | 0.76 | 0.74 | 0.74 | **0.74** | 0.76 | 0.74 | 0.74 | 0.93 | 0.87 | 0.84 | 0.74 | 0.93 | 0.74 | 0.82 | 0.81 | 0.83 | 0.74 | 0.74 | 1.05 | 1.19 |
| **Mean (35)** | 0.50 | 0.50 | 0.49 | 0.49 | **0.48** | 0.51 | 0.50 | 0.49 | 0.61 | 0.57 | 0.54 | 0.49 | 0.56 | 0.50 | 0.63 | 0.61 | 0.63 | 0.49 | 0.50 | 0.67 | 0.97 |

<0.20 0.20–0.40 0.40–0.60 0.60–0.80 0.80–1.00 >1.00

Table 5: Mean NLL by task type for finetuned models (bf16), restricted to the 35 experiments (of 46) that have both a reported cognitive model baseline and a well-defined discrete response space admitting a chance-level baseline $\ln(k)$, where $k$ is the number of response options. 11 experiments lacking either a cognitive baseline or a computable $\ln(k)$ (continuous, mixed, or variable action spaces) are excluded. Finetuned families: `Qwentaur`, `Llama-Centaur`, `Smoltaur`, `Olmotaur`. Base columns: `Q-8B`/`Q-14B` (Qwen3), `L-8B` (Llama-3.1). Subscript $_r$ denotes our reproducing evaluation of the original Centaur model distributed by [binz2025foundation], evaluated under identical python library and CUDA versions as our small foundation models for fair comparison. Subscript $_p$ denotes values published by [binz2025foundation]. $70B_p$ is shown for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. Cell colour encodes performance; **bold+underline** marks the best model, underline the second-best. Full per-experiment results in Appendix **??**.

| | Qwentaur | | | | | Llama-Centaur | | | Smoltaur | | | | Olmotaur | | Base | | | [binz2025foundation] | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Task type** | 0.6B | 1.7B | 4B | 8B | 14B | 1B | 3B | 8B | 0.1B | 0.4B | 1.7B | 3B | 1B | 7B | Q-8B | Q-14B | L-8B | $70B_p$ | $70B_r$ | $Cog._p$ |
| Decision (8) | 0.53 | 0.53 | 0.54 | 0.54 | **0.54** | 0.53 | 0.53 | 0.54 | 0.41 | 0.46 | 0.48 | 0.54 | 0.47 | 0.53 | 0.34 | 0.35 | 0.33 | 0.56 | 0.52 | 0.27 |
| MDP (5) | 0.28 | 0.30 | 0.31 | 0.31 | **0.32** | 0.28 | 0.30 | 0.31 | 0.18 | 0.20 | 0.26 | 0.30 | 0.20 | 0.30 | 0.15 | 0.17 | 0.19 | 0.32 | 0.31 | 0.11 |
| Bandit (12) | 0.48 | 0.48 | 0.49 | 0.49 | **0.50** | 0.46 | 0.48 | 0.49 | 0.36 | 0.41 | 0.44 | 0.49 | 0.42 | 0.48 | 0.35 | 0.36 | 0.33 | 0.46 | 0.45 | 0.39 |
| Memory (4) | 0.57 | 0.57 | 0.58 | 0.58 | 0.58 | 0.56 | 0.57 | 0.58 | 0.46 | 0.51 | 0.55 | 0.57 | 0.53 | 0.57 | 0.42 | 0.47 | 0.43 | 0.58 | **0.58** | 0.36 |
| Misc. (2) | 0.57 | 0.57 | 0.58 | 0.58 | **0.58** | 0.56 | 0.57 | 0.58 | 0.45 | 0.53 | 0.55 | 0.58 | 0.55 | 0.57 | 0.41 | 0.42 | 0.40 | 0.57 | 0.57 | 0.55 |
| Sup. learn. (3) | 0.30 | 0.29 | 0.31 | 0.31 | 0.31 | 0.29 | 0.31 | 0.31 | 0.18 | 0.23 | 0.25 | 0.30 | 0.20 | **0.31** | 0.24 | 0.24 | 0.23 | 0.30 | 0.30 | 0.11 |
| **Mean (34)** | 0.46 | 0.46 | 0.47 | 0.48 | **0.48** | 0.45 | 0.47 | 0.48 | 0.35 | 0.39 | 0.43 | 0.47 | 0.40 | 0.47 | 0.32 | 0.34 | 0.32 | 0.47 | 0.46 | 0.30 |

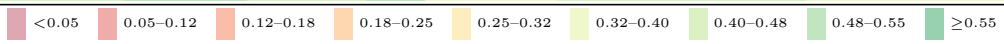
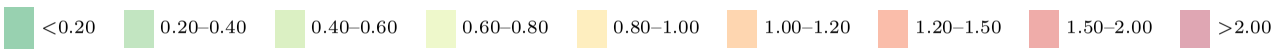



Table 6: Fraction of available information captured above chance, $(\ln k - \text{NLL}) / \ln k$, by task type for finetuned models (bf16). A value of 0 indicates chance-level performance; 1 indicates perfect prediction. Restricted to 34 experiments (of 46) with both a cognitive model baseline and a well-defined discrete response space ($\ln k > 0$); 12 experiments are excluded. Finetuned families: `Qwentaur`, `Llama-Centaur`, `Smoltaur`, `Olmotaur`. Base columns: `Q-8B`/`Q-14B` (Qwen3), `L-8B` (Llama-3.1). Subscript $_r$ denotes our reproducing evaluation of the original Centaur model distributed by [binz2025foundation], evaluated under identical python library and CUDA versions as our small foundation models for fair comparison. Subscript $_p$ denotes values published by [binz2025foundation]. $70B_p$ is shown for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. **Bold+underline** marks the best model, underline the second-best. Full per-experiment results in Appendix **??**.

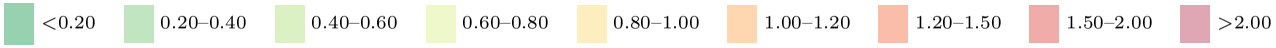
| | | Qwen 2.5/3 family | | | | | | Llama 3/3.1/3.2 family | | | | | | | | | | [binz2025foundation] | **Chance** | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Base | | Qwentaur | | Other cog. | Non-cog. | Base | | Llama-Centaur | | Other cog. | | Non-cog. | | | | | | |
| **Experiment** | **Type** | 8B | 14B | 8B | 14B | Socr.-14B | Herm.-14B | 3B | 8B | 3B | 8B | Be.FM-8B | Socr.-8B | Herm.-3B | Nemo.-4B | Herm.-8B | Nemo.-8B | C-70B$_r$ | Cog.$_p$ | ln($k$) |
| Balloon analog risk task [frey2017risk] | Decision | 0.08 | 0.08 | 0.06 | 0.06 | 5.87 | 0.08 | 0.08 | 0.08 | 0.07 | **0.06** | 0.10 | 13.52 | 0.09 | 0.17 | 0.09 | 0.18 | 0.07 | 0.09 | 0.69 |
| CPC18 [plonsky2018when] | Decision | 0.40 | 0.40 | 0.34 | **0.33** | 8.54 | 0.40 | 0.41 | 0.39 | 0.35 | 0.34 | 0.47 | 14.34 | 0.44 | 0.95 | 0.42 | 0.64 | 0.35 | 0.66 | 0.69 |
| Columbia card task [frey2017risk] | Decision | 0.25 | 0.25 | 0.20 | 0.20 | 6.72 | 0.26 | 0.27 | 0.25 | 0.20 | **0.20** | 0.26 | 14.07 | 0.27 | 0.48 | 0.26 | 0.49 | 0.21 | 0.26 | 0.69 |
| Decisions from description [wulff2018sampling] | Decision | 0.99 | 1.01 | 0.61 | 0.60 | 10.13 | 1.02 | 1.20 | 0.96 | 0.63 | 0.60 | 1.28 | 13.59 | 1.34 | 2.99 | 1.17 | 4.44 | **0.59** | 0.61 | 0.69 |
| Experiential-symbolic task [garcia2023experiential] | Decision | 0.64 | 0.57 | 0.46 | 0.46 | 7.86 | 0.58 | 0.69 | 0.64 | 0.47 | 0.46 | 0.73 | 12.95 | 0.79 | 1.37 | 0.67 | 0.95 | **0.46** | — | — |
| Gardening task [flesch2018comparing] | Decision | 0.47 | 0.45 | 0.38 | 0.38 | 10.28 | 0.45 | 0.64 | 0.48 | 0.39 | **0.38** | 0.48 | 16.26 | 0.53 | 0.61 | 0.46 | 0.64 | 0.49 | 0.91 | 0.69 |
| Go/no-go [enkavi2019large] | Decision | 0.01 | 0.00 | **0.00** | **0.00** | 4.24 | 0.00 | 0.01 | 0.00 | **0.00** | **0.00** | 0.01 | 12.55 | 0.01 | 0.04 | 0.00 | 0.01 | **0.00** | 0.08 | 0.00 |
| Intertemporal choice [ruggeri2022globalizability] | Decision | 0.69 | 0.70 | 0.44 | 0.44 | 12.23 | 0.74 | 0.75 | 0.76 | 0.44 | 0.44 | 0.76 | 13.82 | 0.72 | 2.75 | 1.04 | 2.72 | **0.44** | 0.66 | 0.69 |
| Multi-attribute DM [hilbig2014generalized] | Decision | 0.19 | 0.19 | 0.08 | 0.08 | 10.45 | 0.20 | 0.45 | 0.23 | 0.08 | 0.08 | 0.30 | 13.86 | 0.45 | 0.61 | 0.23 | 0.54 | **0.06** | 0.19 | 0.69 |
| Risky choice [krueger2024identifying] | Decision | 0.74 | 0.67 | 0.43 | **0.42** | 7.17 | 0.67 | 0.85 | 0.76 | 0.46 | 0.43 | 0.85 | 16.10 | 0.94 | 1.79 | 0.82 | 1.35 | 0.43 | — | — |
| choices13k [peterson2021using] | Decision | 0.59 | 0.55 | 0.43 | **0.43** | 9.40 | 0.53 | 0.59 | 0.55 | 0.44 | 0.43 | 0.68 | 13.44 | 0.65 | 1.65 | 0.60 | 1.13 | 0.43 | 0.66 | 0.69 |
| Multi-task RL [tomov2021multitask] | MDP | 0.70 | 0.68 | 0.58 | **0.57** | 7.95 | 0.66 | 0.70 | 0.66 | 0.59 | 0.57 | 0.82 | 14.70 | 0.81 | 1.20 | 0.70 | 1.06 | 0.57 | 1.04 | 1.10 |
| Tile-revealing task [kumar2023disentangling] | MDP | 2.73 | 2.60 | 0.07 | 0.00 | 35.15 | 2.62 | 3.98 | 2.92 | 0.00 | **0.00** | 3.03 | 65.75 | 2.93 | 3.48 | 2.85 | 3.37 | 2.29 | — | — |
| Two-step task [kool2017cost] | MDP | 0.65 | 0.63 | 0.53 | **0.53** | 7.90 | 0.63 | 0.62 | 0.61 | 0.53 | **0.53** | 0.71 | 14.06 | 0.67 | 1.17 | 0.67 | 0.81 | 0.53 | 0.60 | 0.69 |
| Two-step task [kool2016when] | MDP | 0.65 | 0.63 | 0.49 | 0.48 | 8.82 | 0.63 | 0.62 | 0.61 | 0.50 | 0.49 | 0.72 | 14.63 | 0.68 | 1.15 | 0.67 | 0.82 | **0.48** | 0.60 | 0.69 |
| Two-step task [zorowitz2023data] | MDP | 0.61 | 0.59 | **0.51** | 0.51 | 7.68 | 0.58 | 0.60 | 0.58 | 0.51 | 0.51 | 0.65 | 13.18 | 0.63 | 1.03 | 0.62 | 0.82 | 0.51 | 0.60 | 0.69 |
| Virtual subway network [tomov2020discovery] | MDP | 1.55 | 1.57 | 1.13 | **1.12** | 11.39 | 1.60 | 1.60 | 1.53 | 1.16 | 1.13 | 1.61 | 18.21 | 1.69 | 2.41 | 1.64 | 2.10 | 1.16 | — | 1.61 |
| Zoopermarket [ludwig2023human] | MDP | 0.60 | 0.59 | 0.50 | **0.48** | 6.62 | 0.59 | 0.62 | 0.60 | 0.52 | 0.50 | 0.62 | 13.75 | 0.65 | 1.49 | 0.63 | 1.09 | 0.50 | 0.60 | 0.69 |
| Cond. assoc. learning [collins2014working] | Memory | 0.69 | 0.61 | 0.52 | **0.51** | 10.04 | 0.59 | 0.73 | 0.70 | 0.53 | 0.52 | 0.72 | 16.16 | 0.80 | 0.92 | 0.68 | 0.95 | 0.52 | 0.86 | 1.10 |
| Digit span [enkavi2019large] | Memory | 0.70 | 0.69 | 0.57 | **0.57** | 2.96 | 0.70 | 0.84 | 0.75 | 0.59 | 0.58 | 0.83 | 4.65 | 0.87 | 2.16 | 0.83 | 1.02 | 0.57 | 0.94 | 2.30 |
| Episodic long-term memory [popov2023intent] | Memory | 1.30 | 1.18 | 0.94 | 0.90 | 9.08 | 1.21 | 1.65 | 1.43 | 1.02 | 0.92 | 1.52 | 17.12 | 1.82 | 3.07 | 1.72 | 2.22 | **0.89** | — | 0.69 |
| N-back [enkavi2019large] | Memory | 0.55 | 0.53 | 0.41 | 0.40 | 9.24 | 0.53 | 0.57 | 0.57 | 0.42 | 0.41 | 0.57 | 14.42 | 0.59 | 0.79 | 0.57 | 0.75 | **0.40** | 0.58 | 0.69 |
| Recall and recognition [cox2018information] | Memory | 1.41 | 1.36 | 1.12 | 1.09 | 9.33 | 1.38 | 1.56 | 1.37 | 1.14 | 1.10 | 1.53 | 24.31 | 1.65 | 4.16 | 1.61 | 2.60 | **1.07** | — | — |
| Recent probes [enkavi2019large] | Memory | 0.41 | 0.35 | 0.26 | 0.26 | 9.92 | 0.34 | 0.56 | 0.34 | 0.26 | **0.26** | 0.40 | 14.16 | 0.61 | 1.00 | 0.39 | 1.06 | 0.26 | 0.39 | 0.69 |




**Bold+underline** = best; underline = second-best. Lower is better. "—" = not available. Models are grouped by base-model family. "Base" = pretrained model without finetuning (bf16); "Other cog." = other cognitive/behavioural foundation models finetuned on non-Psych-101 behavioural data; "Non-cog." = models finetuned for general instruction-following or reasoning. `Herm.` = Hermes [teknium2024hermes]; `Nemo.` = Nemotron [nvidia2025nemotron]; `Socr.` = Socrates [kolluri2025finetuning]; `Be.FM` = [xie2025fm]. Base model versions differ within families: `Qwentaur` uses Qwen3; `Socr.-14B` uses Qwen2.5; `Herm.-14B` uses Qwen2.5. `Llama-Centaur` uses Llama-3.1/3.2; `Socr.-8B` uses Llama-3; `Be.FM` uses Llama-3.1.

Table 7: Comparison of cognitively finetuned models with other behavioural foundation models and non-cognitive controls on Psych-101, grouped by base-model family. Within each family, `Qwentaur`/`Llama-Centaur` (cognitive SFT on Psych-101) consistently outperform both other behavioural FMs and non-cognitive controls, demonstrating that alignment with human behavioural patterns requires cognitive finetuning on structured experimental data, not merely finetuning per se.

| | | Qwen 2.5/3 family | | | | | | Llama 3/3.1/3.2 family | | | | | | | | | | [binz2025foundation] | **Chance** | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Base | | Qwentaur | | Other cog. | Non-cog. | Base | | Llama-Centaur | | Other cog. | | Non-cog. | | | | | | |
| **Experiment** | **Type** | 8B | 14B | 8B | 14B | Socr.-14B | Herm.-14B | 3B | 8B | 3B | 8B | Be.FM-8B | Socr.-8B | Herm.-3B | Nemo.-4B | Herm.-8B | Nemo.-8B | C-70B$_r$ | Cog.$_p$ | ln($k$) |
| Grammar judgement [jansen2021rational] | Misc. | 1.89 | 1.91 | 1.44 | 1.43 | 10.37 | 1.94 | 2.16 | 2.12 | 1.45 | 1.44 | 2.11 | 12.88 | 2.27 | 3.90 | 2.53 | 3.12 | 1.44 | **1.41** | — |
| Probabilistic reasoning [zhu2020bayesian] | Misc. | 2.57 | 2.58 | 2.35 | **2.34** | 2.92 | 2.59 | 2.80 | 2.71 | 2.47 | 2.43 | 2.84 | 3.40 | 2.98 | 4.05 | 2.97 | 3.23 | 2.40 | — | — |
| Serial reaction time task [wu2023chunking] | Misc. | 0.19 | 0.19 | 0.17 | **0.17** | 5.09 | 0.19 | 0.23 | 0.20 | 0.17 | 0.17 | 0.25 | 14.43 | 0.21 | 0.25 | 0.20 | 0.26 | 0.18 | 0.20 | 1.39 |
| THINGS odd-one-out [hebart2023things] | Misc. | 1.15 | 1.13 | 0.79 | **0.79** | 12.15 | 1.15 | 1.18 | 1.16 | 0.82 | 0.79 | 1.19 | 14.31 | 1.24 | 2.13 | 1.28 | 1.75 | 0.80 | 0.83 | 1.10 |
| Changing bandit [xiong2023neural] | Bandit | 0.35 | 0.34 | 0.29 | **0.29** | 7.52 | 0.34 | 0.90 | 0.39 | 0.31 | 0.29 | 0.42 | 14.76 | 0.39 | 0.78 | 0.36 | 0.53 | 0.45 | 0.44 | 0.69 |
| Decisions from experience [wulff2018sampling] | Bandit | 0.78 | 0.61 | 0.49 | 0.48 | 7.18 | 0.64 | 0.71 | 0.65 | 0.49 | 0.48 | 0.83 | 13.74 | 0.69 | 2.29 | 0.88 | 1.78 | **0.47** | 0.54 | — |
| Drifting four-armed bandit [bahrami2020four] | Bandit | 0.91 | 0.91 | 0.71 | **0.71** | 8.43 | 0.90 | 0.94 | 0.90 | 0.72 | 0.71 | 0.99 | 13.91 | 0.99 | 2.10 | 0.96 | 1.78 | 0.71 | 0.90 | 1.39 |
| Horizon task [feng2021dynamics] | Bandit | 0.36 | 0.34 | 0.22 | 0.22 | 7.55 | 0.35 | 0.67 | 0.44 | 0.24 | **0.22** | 0.39 | 15.10 | 0.43 | 0.65 | 0.36 | 0.62 | 0.40 | 0.36 | 0.69 |
| Horizon task [sadeghiyeh2020temporal] | Bandit | 0.65 | 0.65 | 0.58 | 0.58 | 8.62 | 0.66 | 0.68 | 0.66 | 0.58 | 0.58 | 0.69 | 14.94 | 0.73 | 1.10 | 0.69 | 0.93 | 0.58 | **0.36** | 0.69 |
| Horizon task [somerville2017charting] | Bandit | 0.47 | 0.49 | 0.34 | **0.33** | 8.49 | 0.49 | 0.57 | 0.51 | 0.34 | 0.34 | 0.53 | 15.18 | 0.57 | 0.86 | 0.47 | 0.74 | 0.35 | 0.36 | 0.69 |
| Horizon task [waltz2020differential] | Bandit | 0.27 | 0.24 | 0.15 | 0.15 | 7.82 | 0.26 | 0.37 | 0.30 | 0.16 | **0.14** | 0.33 | 15.42 | 0.35 | 0.54 | 0.26 | 0.54 | 0.15 | 0.36 | 0.69 |
| Horizon task [wilson2014humans] | Bandit | 0.53 | 0.54 | 0.45 | 0.45 | 8.06 | 0.54 | 0.62 | 0.55 | 0.46 | 0.45 | 0.57 | 15.39 | 0.62 | 0.89 | 0.56 | 0.79 | 0.48 | **0.36** | 0.69 |
| Iowa gambling task [steingroever2015data] | Bandit | 1.07 | 1.07 | 0.91 | 0.91 | 8.45 | 1.06 | 1.06 | 1.04 | 0.92 | 0.91 | 1.11 | 13.23 | 1.13 | 2.27 | 1.09 | 1.60 | **0.91** | 1.16 | 1.39 |
| Prob. instrumental learning [lefebvre2017behavioural] | Bandit | 0.59 | 0.56 | **0.49** | 0.49 | 8.30 | 0.53 | 0.58 | 0.56 | 0.50 | 0.49 | 0.61 | 15.92 | 0.68 | 1.16 | 0.54 | 0.86 | 0.50 | 0.50 | 0.69 |
| Spatially correlated MAB [wu2018generalization] | Bandit | 2.44 | 2.46 | 1.87 | 1.86 | 3.14 | 2.48 | 2.64 | 2.58 | 2.00 | 1.93 | 2.82 | 5.43 | 2.64 | 3.93 | 2.78 | 3.19 | **1.82** | 2.76 | 3.40 |
| Structured bandit [schulz2020finding] | Bandit | 0.87 | 0.83 | 0.65 | **0.64** | 1.57 | 0.80 | 0.89 | 0.85 | 0.66 | 0.65 | 0.98 | 2.12 | 0.91 | 1.41 | 0.90 | 1.08 | 0.64 | 1.05 | 2.08 |
| Two-armed bandit [gershman2018deconstructing] | Bandit | 0.39 | 0.39 | 0.29 | **0.29** | 7.30 | 0.41 | 0.43 | 0.40 | 0.29 | 0.29 | 0.46 | 13.99 | 0.47 | 0.98 | 0.44 | 0.73 | 0.30 | 0.42 | 0.69 |
| Aversive learning [wise2019computational] | Sup. learn. | 4.89 | 4.75 | 4.24 | 3.81 | 7.19 | 4.78 | 5.52 | 5.25 | 4.80 | **3.66** | 5.49 | 18.42 | 5.52 | 7.03 | 5.33 | 6.14 | 4.13 | — | — |
| Medin categorization [levering2020revisiting] | Sup. learn. | 0.59 | 0.55 | 0.50 | **0.50** | 11.00 | 0.56 | 0.60 | 0.57 | 0.50 | 0.50 | 0.63 | 13.53 | 0.63 | 1.15 | 0.62 | 0.90 | 0.50 | 0.53 | — |
| Multiple-cue judgment [collsioo2023numerical] | Sup. learn. | 1.30 | 1.26 | 1.14 | 1.13 | 6.92 | 1.27 | 1.50 | 1.32 | 1.15 | **1.12** | 1.38 | 18.06 | 1.51 | 2.56 | 1.37 | 1.85 | 1.14 | 1.92 | 2.20 |
| Shepard categorization [badham2017deficits] | Sup. learn. | 0.58 | 0.59 | 0.53 | **0.53** | 10.13 | 0.59 | 0.62 | 0.60 | 0.54 | 0.53 | 0.60 | 12.66 | 0.64 | 0.92 | 0.63 | 0.76 | 0.54 | 0.61 | 0.69 |
| Weather prediction task [speekenbrink2008learning] | Sup. learn. | 0.58 | 0.58 | 0.55 | 0.55 | 10.65 | 0.56 | 0.59 | 0.58 | **0.54** | 0.56 | 0.62 | 17.01 | 0.63 | 0.89 | 0.61 | 0.77 | 0.56 | 0.63 | 0.69 |
| **Mean (all 46)** | | 0.89 | 0.87 | 0.66 | **0.64** | 8.65 | 0.87 | 1.02 | 0.92 | 0.68 | 0.64 | 0.99 | 15.07 | 1.02 | 1.72 | 0.98 | 1.43 | 0.71 | 0.69 | 0.98 |




**Bold+underline** = best; underline = second-best. Lower is better. "—" = not available. Models are grouped by base-model family. "Base" = pretrained model without finetuning (bf16); "Other cog." = other cognitive/behavioural foundation models finetuned on non-Psych-101 behavioural data; "Non-cog." = models finetuned for general instruction-following or reasoning. Herm. = Hermes [teknium2024hermes]; Nemo. = Nemotron [nvidia2025nemotron]; Socr. = Socrates [kolluri2025finetuning]; Be.FM = [xie2025fm]. Base model versions differ within families: Qwentaur uses Qwen3; Socr.-14B uses Qwen2.5; Herm.-14B uses Qwen2.5. Llama-Centaur uses Llama-3.1/3.2; Socr.-8B uses Llama-3; Be.FM uses Llama-3.1.

Table 7: Comparison of cognitively finetuned models with other behavioural foundation models and non-cognitive controls on Psych-101, grouped by base-model family (continued).